

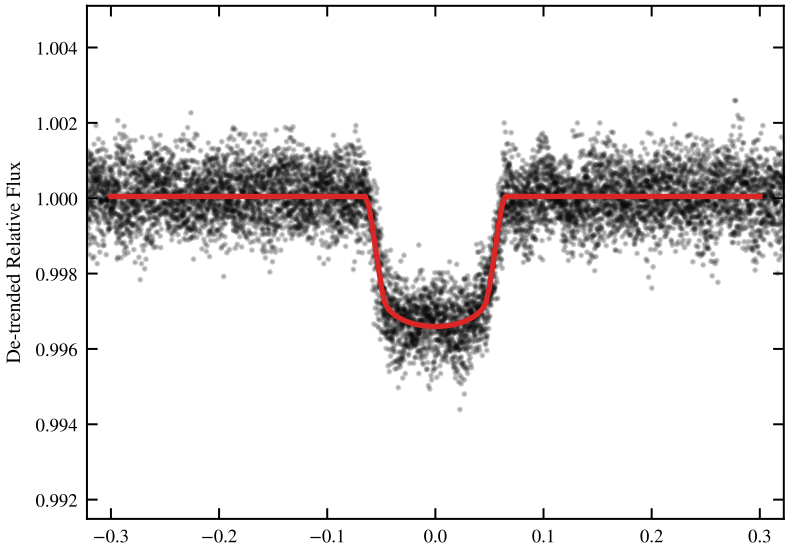
TESS PLANET TRANSIT CHARACTERIZATION REPORT

Target: TIC 409934330 | RA: 302.14773° | Dec: -71.55244° | Sectors: 93, 94

Derived Planetary Parameters

Parameter	Derived Value (MCMC)
Period (d)	2.4888 ± 2.87e-05
Rp/R★	0.058889 ± 0.000565
Rp (R⊕)	12.51 ± 3.75
b	0.7936 ± 0.0157
T14 (h)	3.0929 ± 0.0206
a (AU)	0.039839 ± 0.012
Teq (K)	2712.5 ± 49.8

Primary Phased Transit



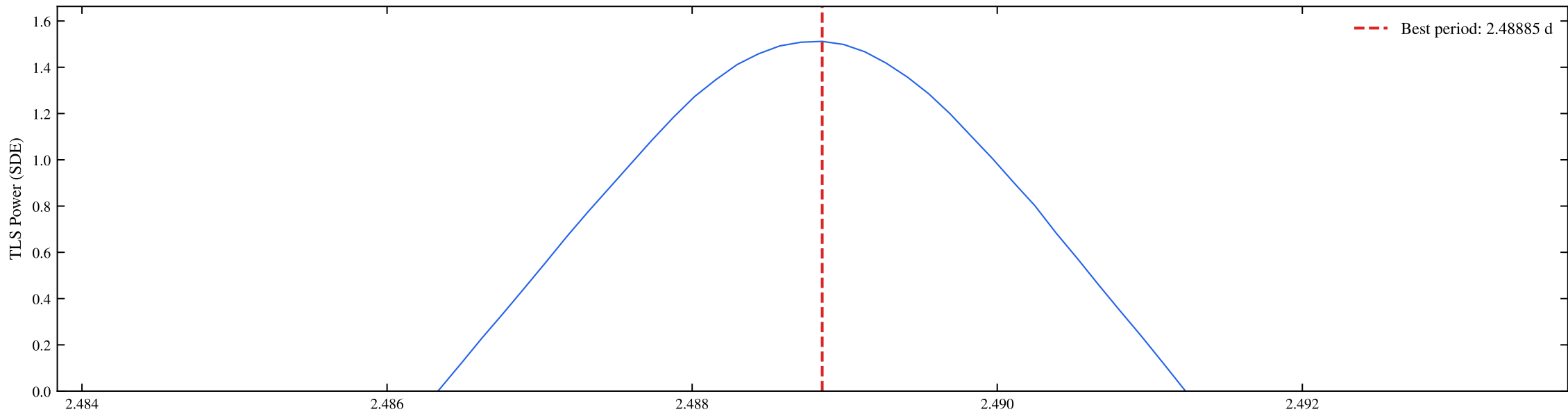
Literature Comparison & MCMC Diagnostics

Parameter / Metric	Literature Value	Derived / Diagnostic
P1 Period (d)	2.48885	2.488803 ± 0.00003
P1 Epoch (BTJD)	N/A	3830.5395 ± 0.0004
Max R-hat (R)	N/A	1.0006
Min ESS	N/A	7164
MCMC Divergences	N/A	0

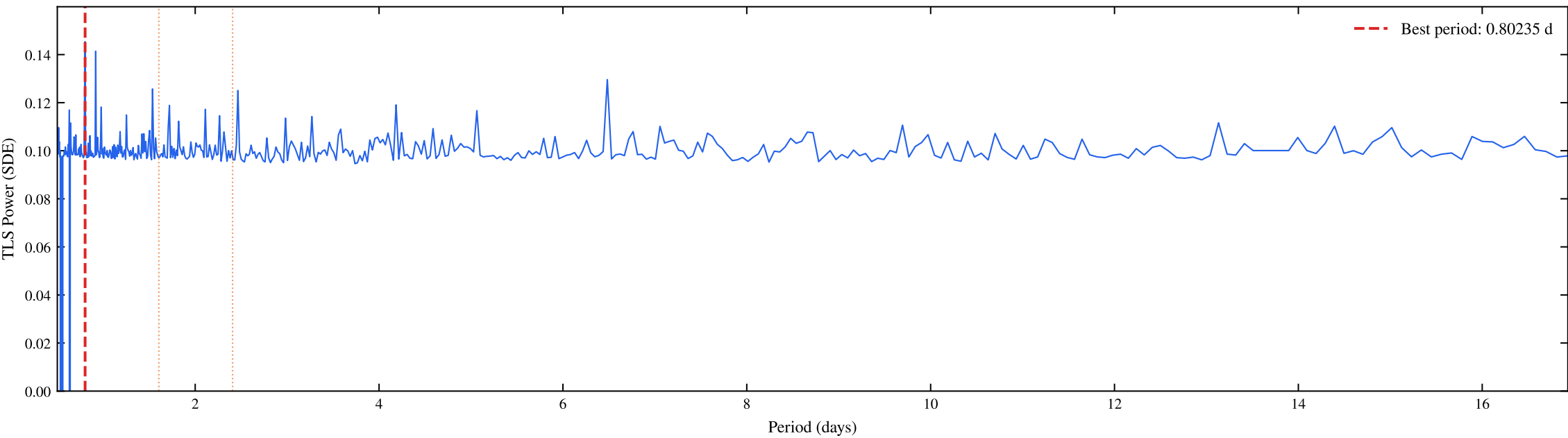
Host Star Properties

Stellar Parameter	Value	Source
R★ (R☉)	1.944 ± 0.583	Derived (Teff scaling)
M★ (M☉)	1.757	Torres 2010
Teff (K)	8048	Gaia DR3
log g (dex)	4.438	Gaia DR3
[Fe/H]	N/A	Catalog
ρ★ (g/cm³)	0.3372	Derived

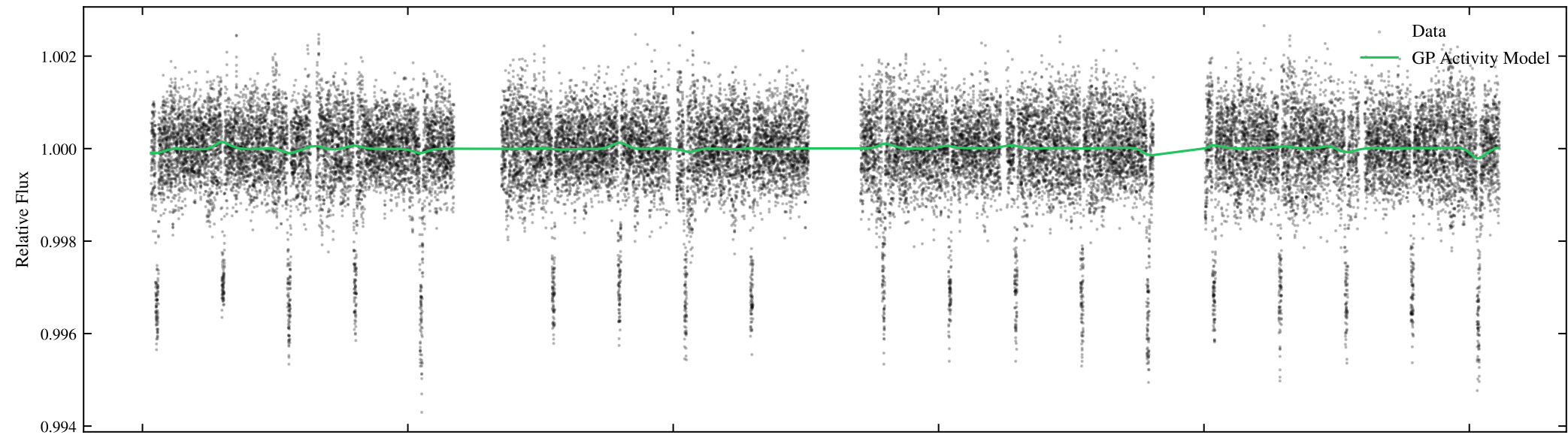
TIC 409934330 | Sectors: 93, 94 | TLS Periodogram (Planet 1 Search) (SDE = 1.51)



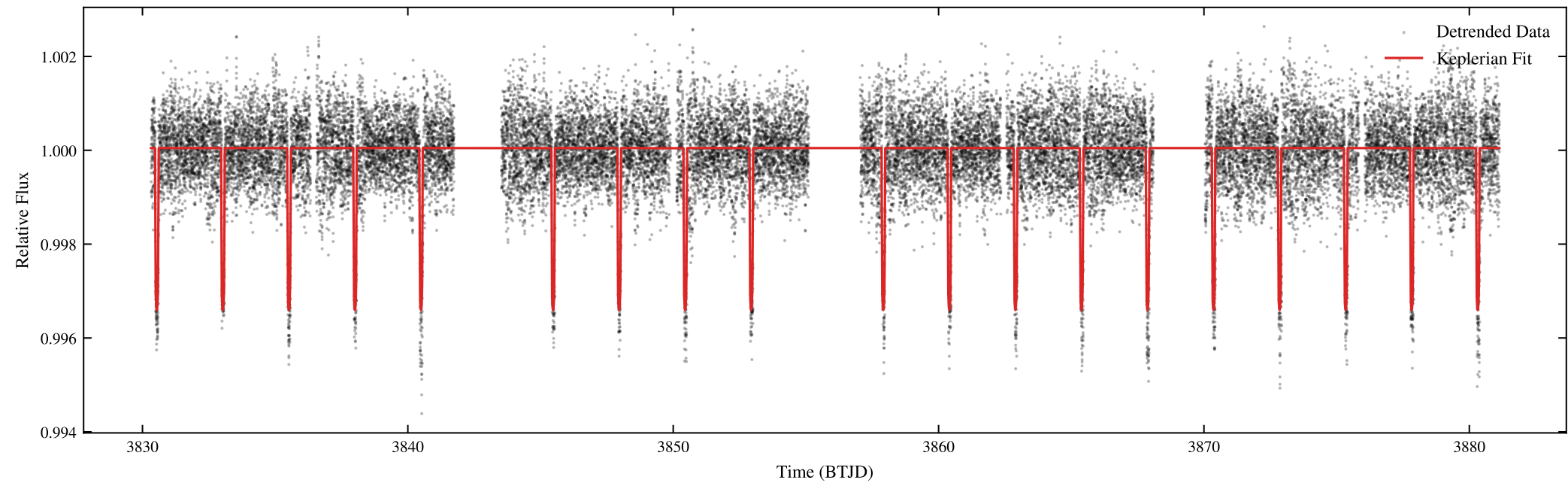
TIC 409934330 | Sectors: 93, 94 | TLS Periodogram (Planet 2 Search) (SDE = 0.15)



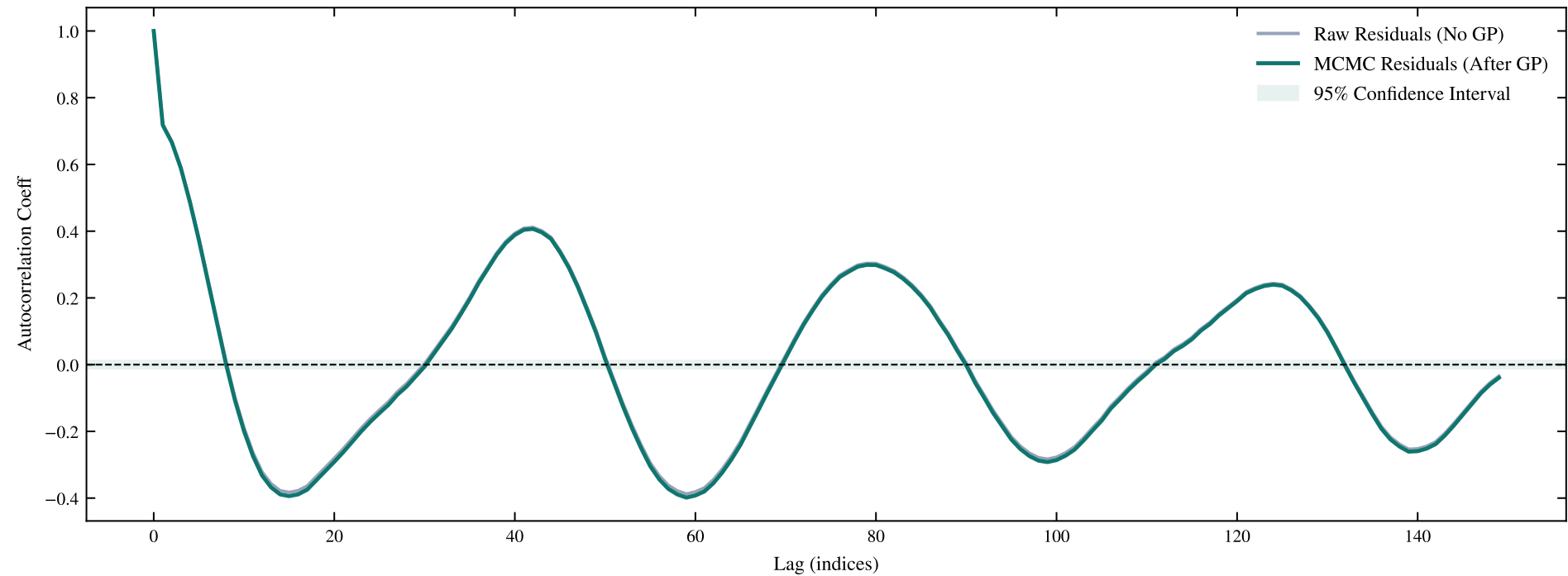
TIC 409934330 | Sectors: 93, 94 | Bayesian GP Detrending (Time Domain)



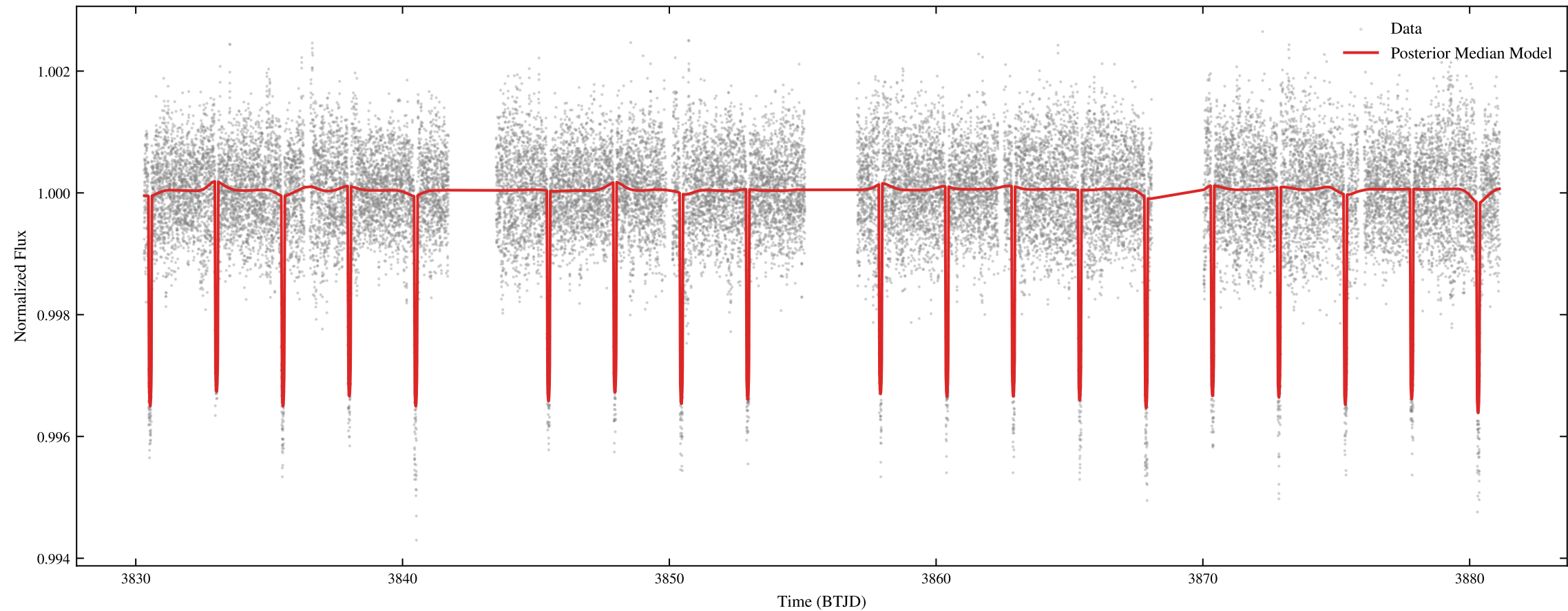
Detrended Light Curve & Best-Fit Transit Model

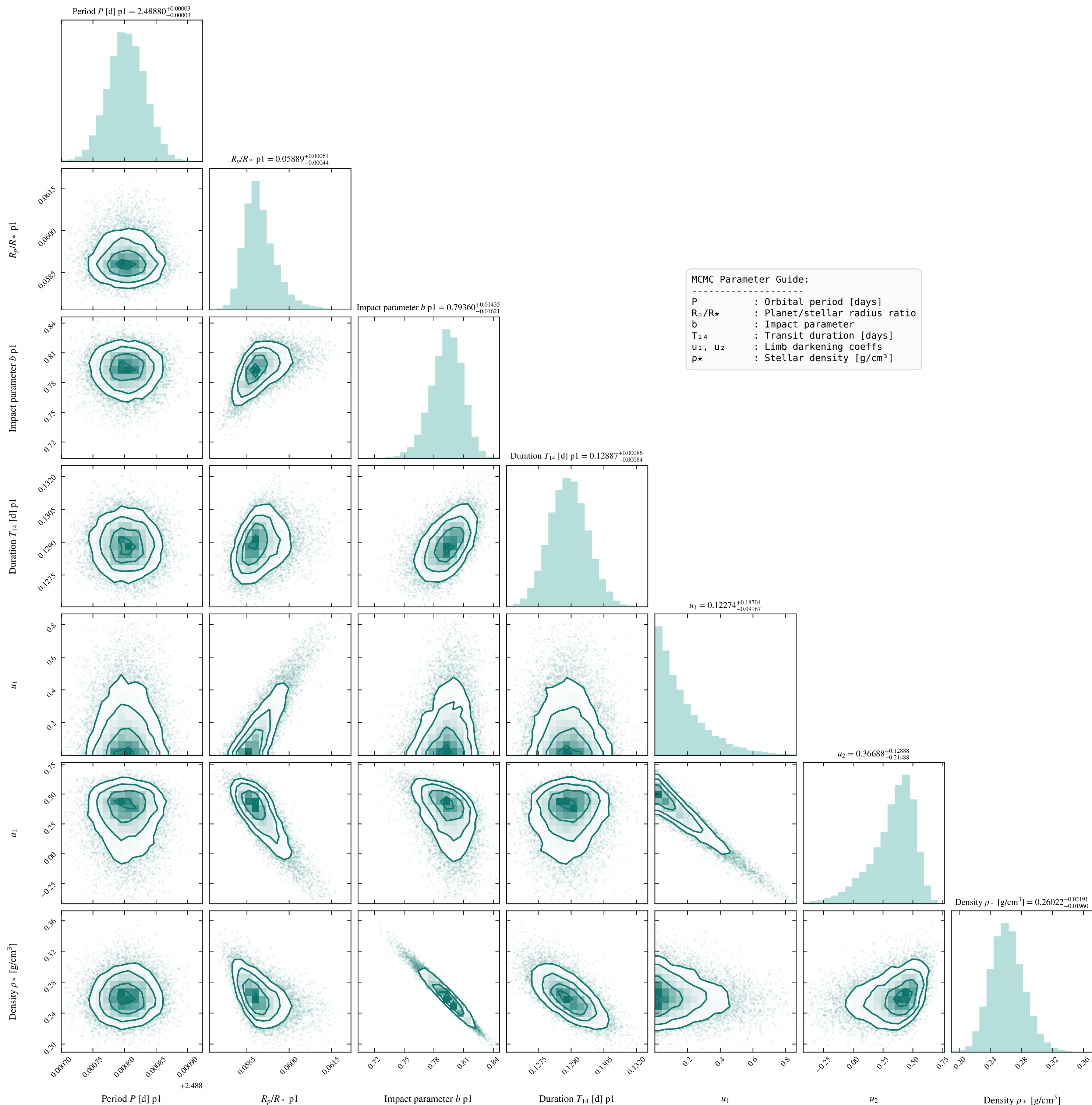


TIC 409934330 | Sectors: 93, 94 | GP Residuals Autocorrelation Function (ACF)

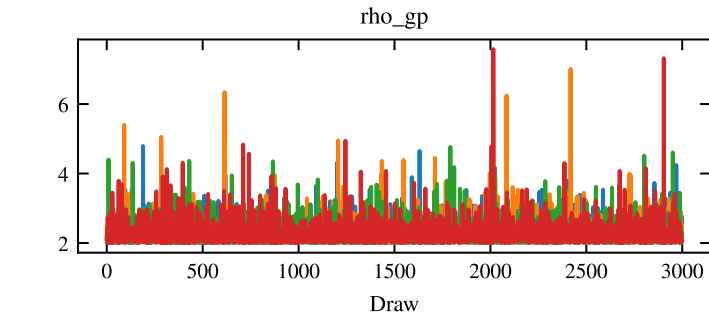
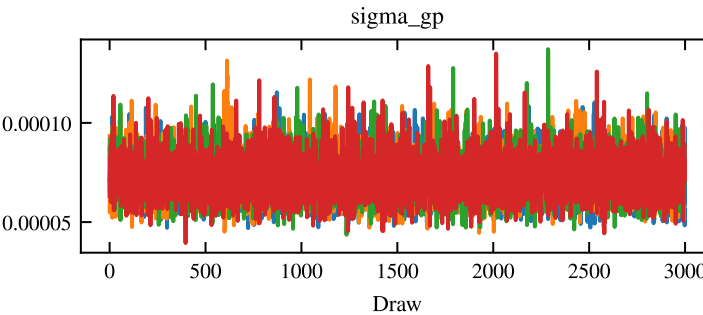
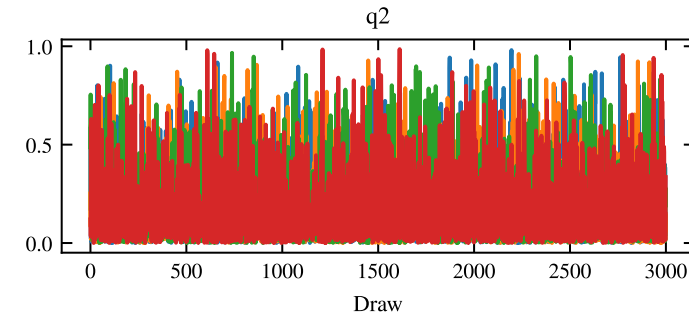
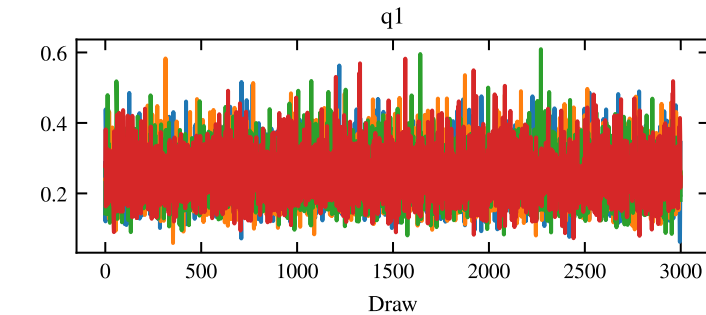
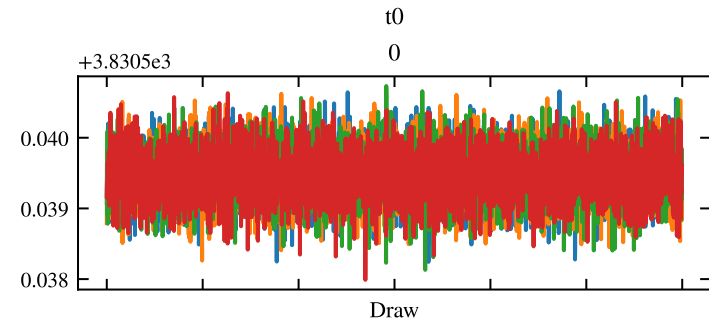
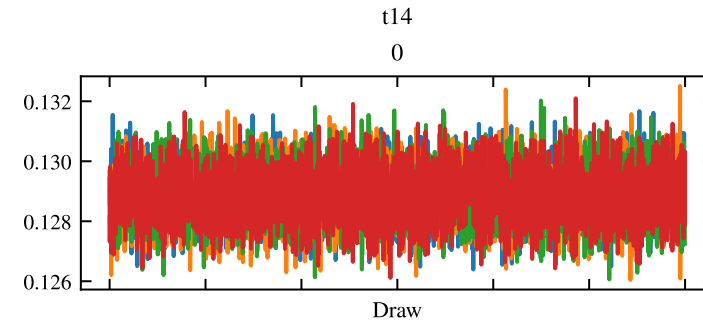
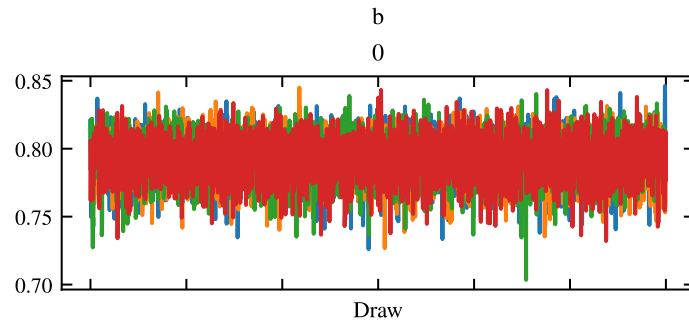
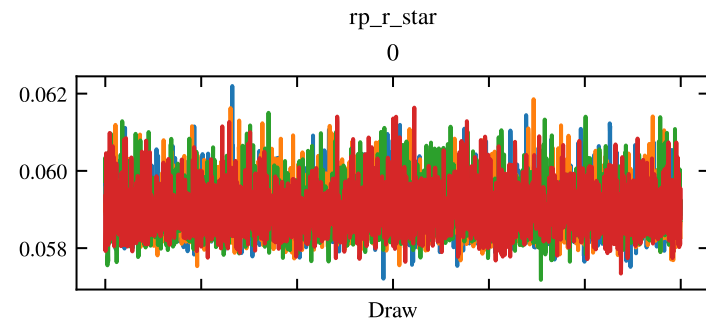


TIC 409934330 | Sectors: 93, 94 | MCMC Posterior Predictive Check

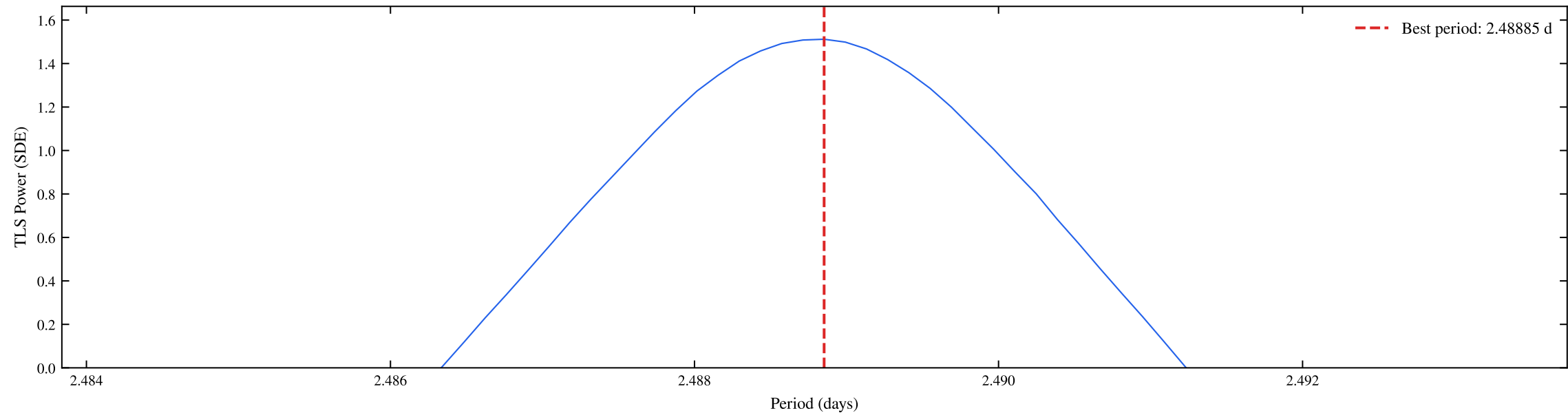




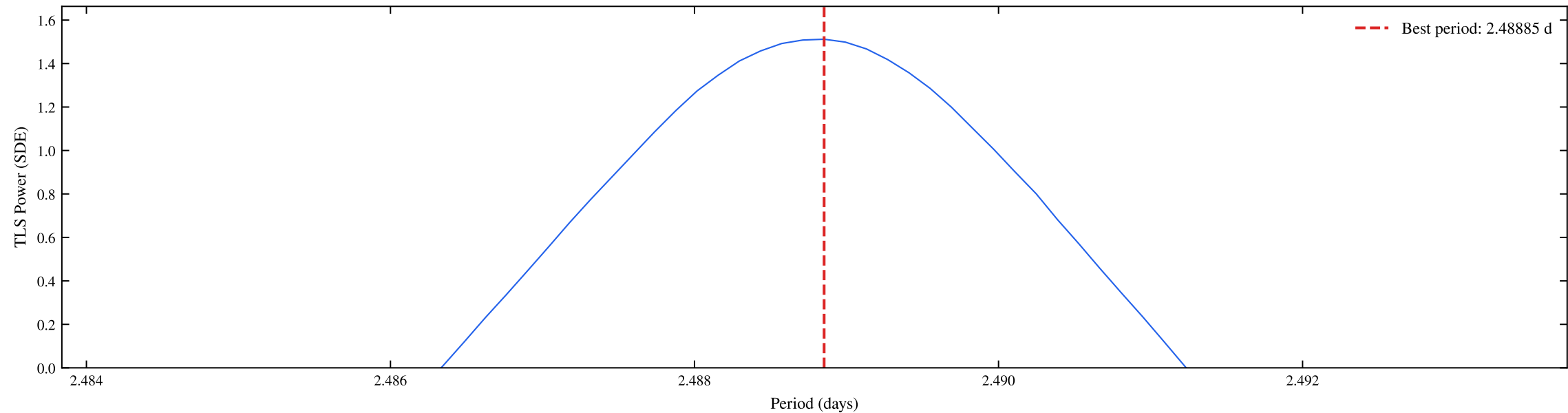
TIC 409934330 | Sectors: 93, 94 | MCMC Trace Plots



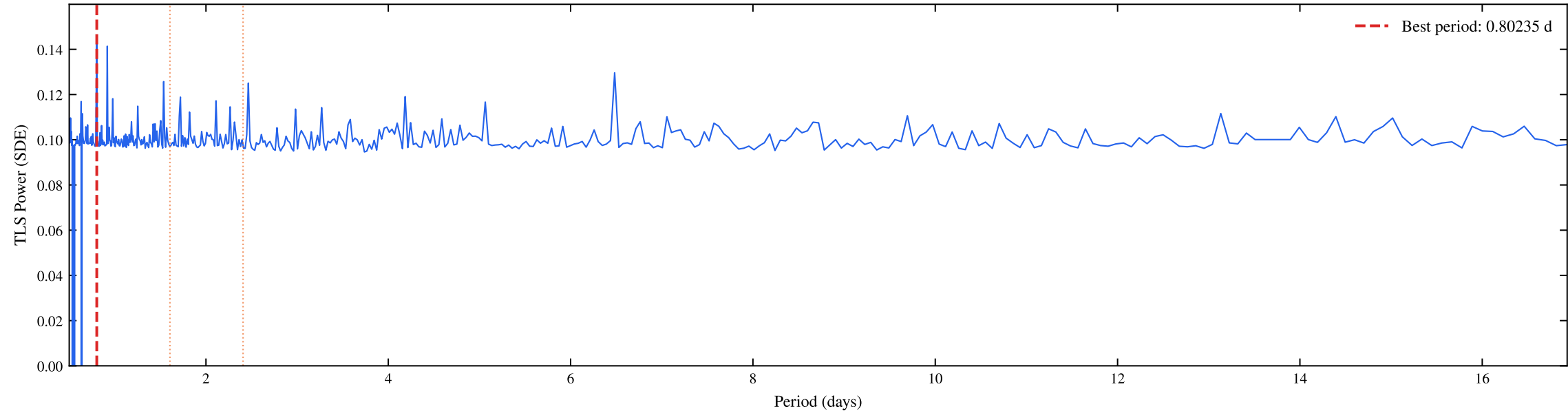
TIC 409934330 | Sectors: 93, 94 | TLS Coarse Periodogram (SDE = 1.51)



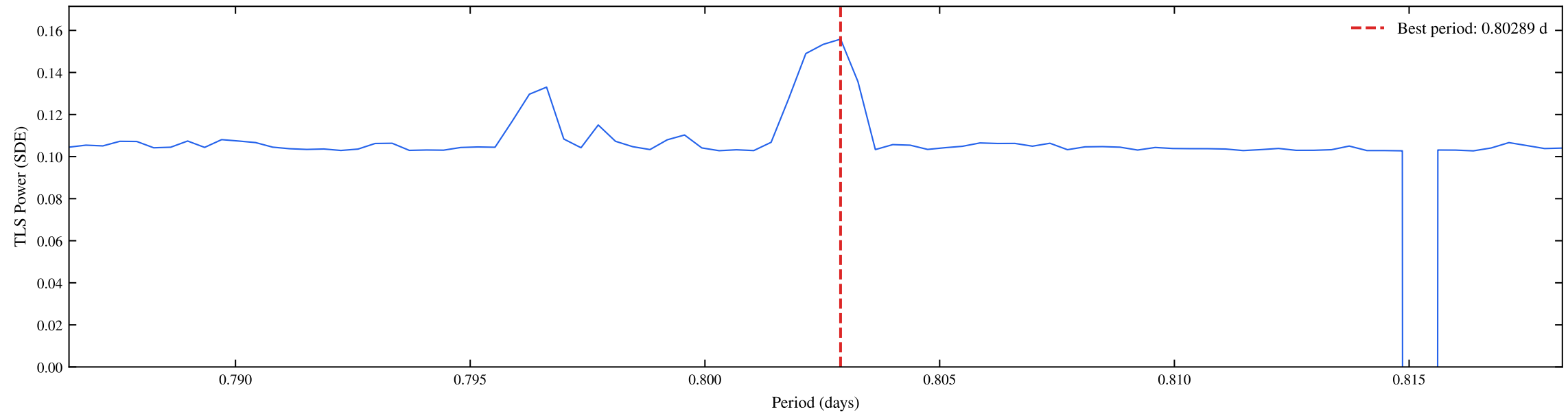
TIC 409934330 | Sectors: 93, 94 | TLS Fine Periodogram (SDE = 1.51)



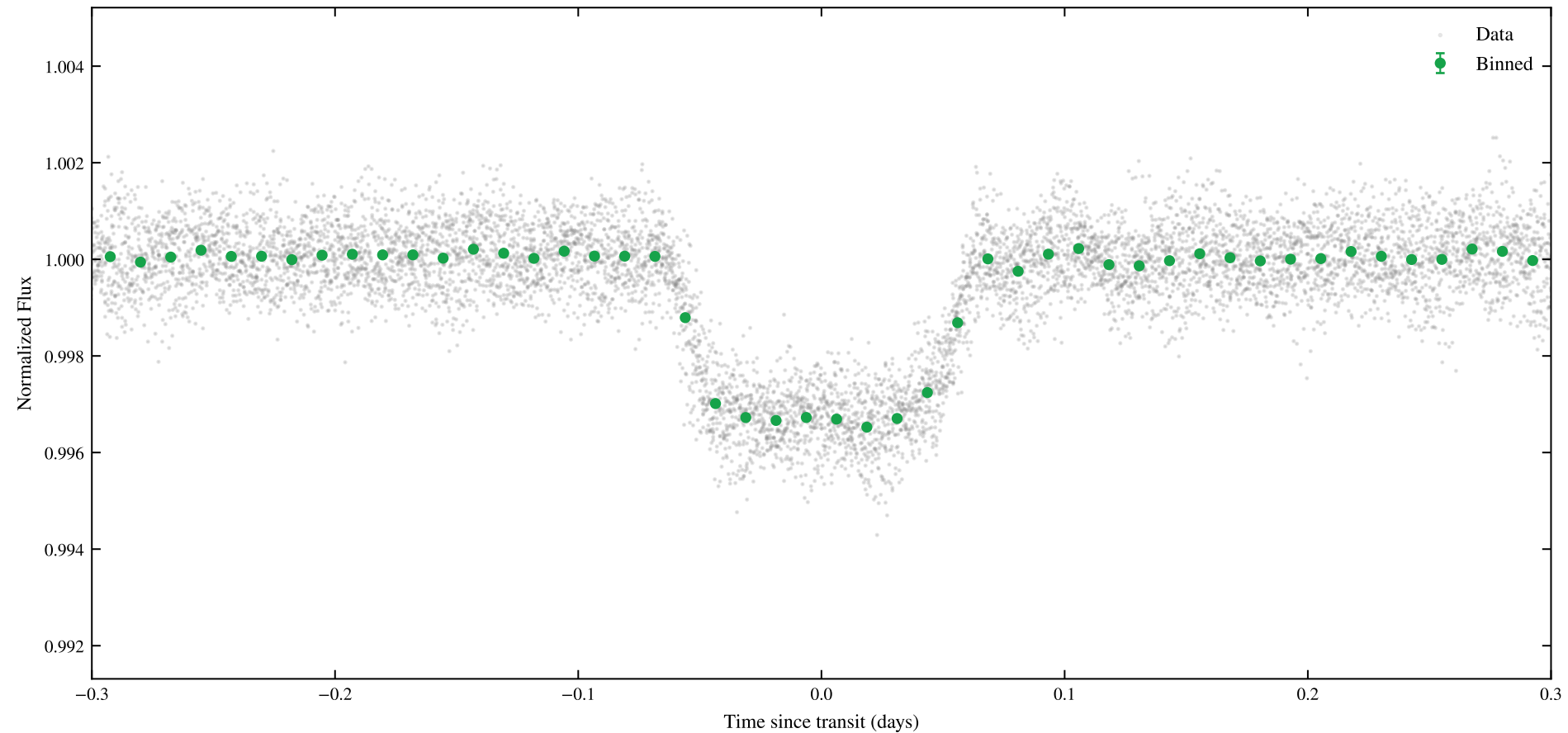
TIC 409934330 | Sectors: 93, 94 | TLS Coarse Periodogram (SDE = 0.15)



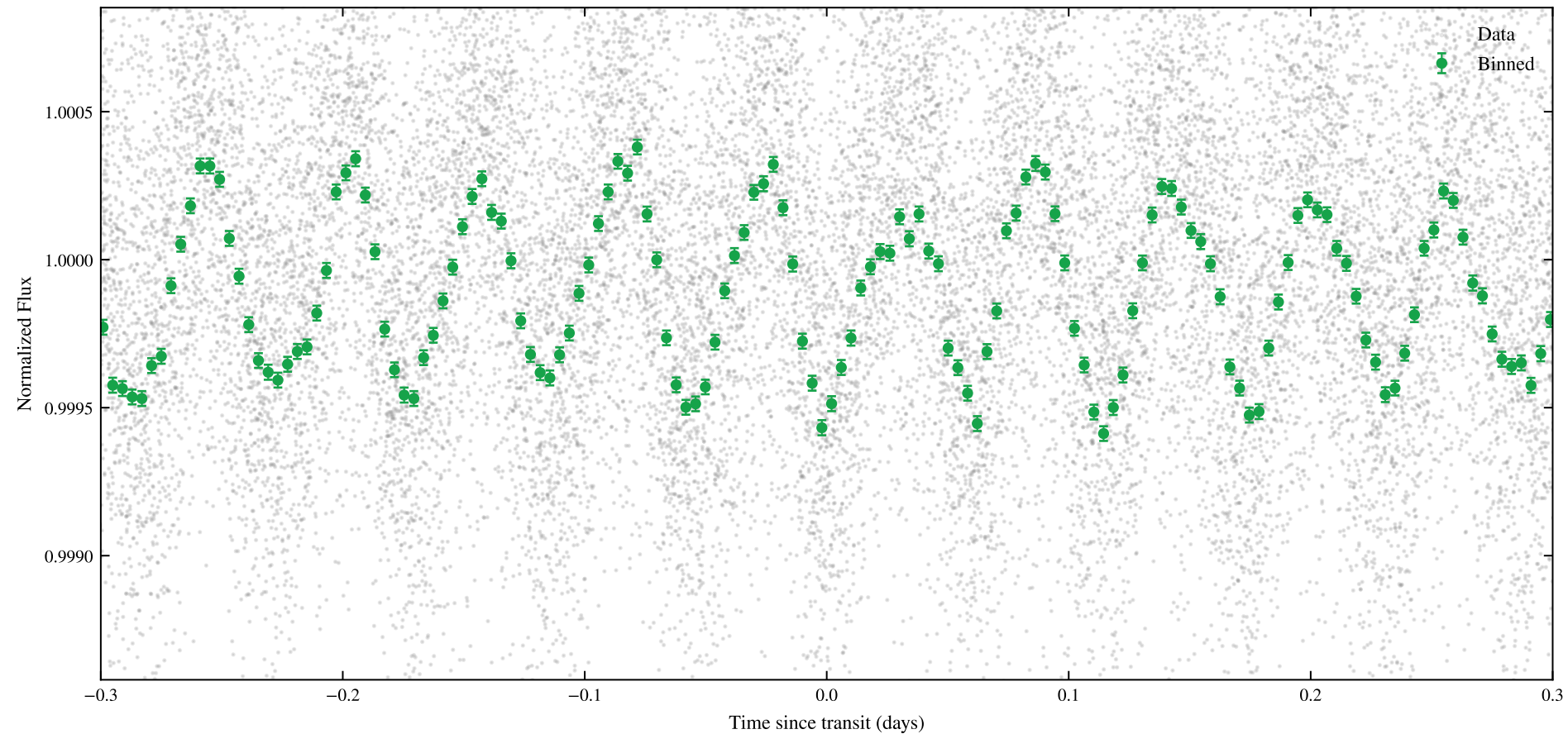
TIC 409934330 | Sectors: 93, 94 | TLS Fine Periodogram (SDE = 0.16)

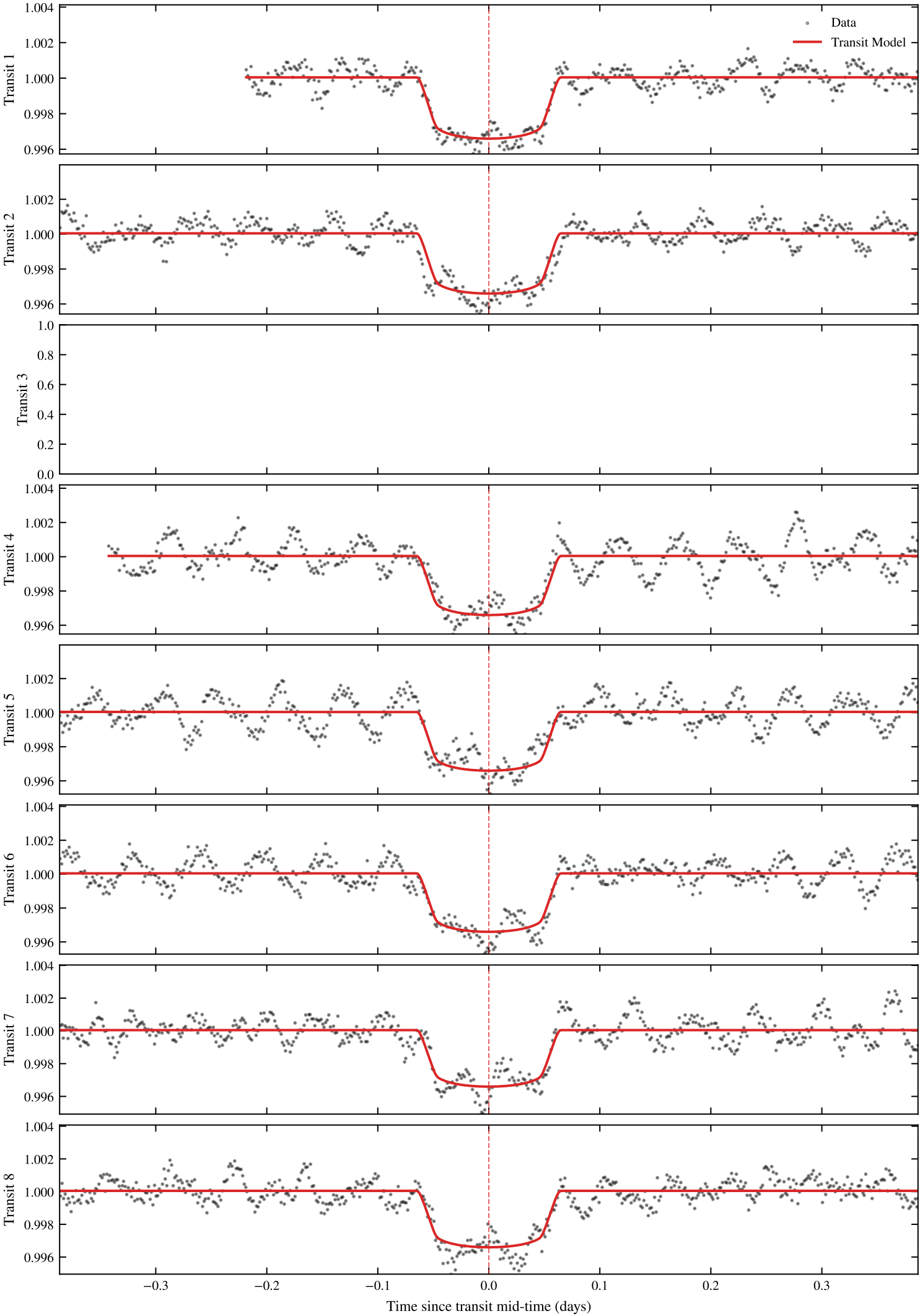


TIC 409934330 | Sectors: 93, 94 | Phase-Folded Transit (P = 2.48885 d)



TIC 409934330 | Sectors: 93, 94 | Phase-Folded Transit (P = 0.80289 d)





TIC 409934330 | Sectors: 93, 94 | Phased MCMC Fit - Planet 1

